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Implementing the reverse acute to chronic workload ratio model to improve movement capacity and roster availability: An example using data from the NFL. International Journal of Strength and Conditioning, 5(1): 1-9, 2025.

Influence of resistance training variables and the Nordic hamstring exercise on biceps femoris architectural adaptations in soccer players: A systematic review. Sports Health, Ahead of Print: 1-10, 2025.

Injury scenarios of ankle sprains in the National Football League: External rotation and eversion mechanisms. The American Journal of Sports Medicine, Ahead of Print: 1-10, 2025.

Introducing the Novel Deceleration Threshold Method: Comparative reliability to previous sprint deceleration analysis methods in team-sport athletes. European Journal of Sport Science, 25: e12278, 2025.

Is it all in your head? Placebo effects in concussion prevention. Sports Medicine, Ahead of Print: 1-17, 2025.

Less pain, same gain? A randomized controlled trial comparing assisted versus non-assisted Nordic hamstring exercise training in elite youth football players. Physical Therapy in Sport, 73: 39-47, 2025.

National Football League players have higher rates of knee extensor mechanism tears during short and normal rest weeks versus long rest weeks. The Journal of Knee Surgery, Ahead of Print: 1-17, 2025.

Nutritional strategies to improve post-exercise recovery and subsequent exercise performance: A narrative review. Sports Medicine, Ahead of Print: 1-19, 2025.

Post-operative rehabilitation and return to sport criteria following distal biceps tendon rupture surgery. JSES International, Ahead of Print: 1-22, 2025.

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Sprint running mechanics are associated with hamstring strain injury: A 6-month prospective cohort study of 126 elite male footballers. British Journal of Sports Medicine, Ahead of Print: 1-8, 2025.

Use of Ankle-Go to assess and predict return to sport after lateral ankle reconstruction for chronic ankle instability. The Orthopaedic Journal of Sports Medicine, 13(3): 1-9, 2025.

Validation of a composite outcome score for assessing return to sports after Achilles tendon repair. The American Journal of Sports Medicine, Ahead of Print: 1-9, 2025.

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A narrative review of the impact of sleep on athletes: Sleep restriction causes and consequences, monitoring, and interventions. Cureus, 16(12): e76635, 2025.

Abstract - This narrative review synthesizes evidence on the impact of sleep on athletic performance, the prevalence and causes of sleep disturbances, and effective monitoring and intervention strategies to enhance sleep quality and duration. A comprehensive review of case studies, observational studies, randomized controlled trials, systematic reviews, and meta-analyses was conducted using PubMed, Cochrane Library, and Google Scholar up to July 2024. Sleep plays a crucial role in the overall well-being and performance of athletes, yet sleep issues are highly prevalent due to factors such as competition schedules, psychological stress, and travel across time zones. Inadequate sleep negatively affects physical strength, endurance, cognitive function, and injury risk. Sleep tracking is highlighted as both a diagnostic and therapeutic tool, with interventions such as extending sleep duration, adopting sleep hygiene practices, and, when necessary, pharmacological approaches proving effective. The review concludes that tailored sleep strategies are essential to optimize athletic performance, recovery, and overall well-being.

Application of the British Athletics Injury Classification in collegiate football athletes: A retrospective, observational study. Sports Health, Ahead of Print: 1-8, 2025.

Background: Lower extremity muscle tears are common sports injuries. The British Athletics Muscle Injury Classification (BAMIC) may provide clinical guidance for tears although its value in American football and college athletes is unknown.

Hypothesis: Clinical outcomes, specifically time to return to play (RTP) (TRTP), in college American football athletes with activity-related hamstring (HS) and quadriceps (QD) tears will be associated with BAMIC.

Study design: Retrospective cohort study.

Level of evidence: Level 3.

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Methods: American football college athletes who sustained a HS or QD tear and underwent magnetic resonance imaging within 7 days at 1 institution during the 2023 season were included. TRTP and reinjury rate (RIR) for HS and QD tears classified by BAMIC and injury characteristics were evaluated.

Results: Of the 24 HS and 10 QD tears in 21 (17.3%) of 121 athletes (20 ± 1.2 years; 100.0% male), most (60.0%) were BAMIC 2a-3b. Most (89.3%) injuries occurred before the season, and all occurred in practice. Overall median TRTP was 26 days (interquartile range, 17.8-33.0), translating to about 4 weeks missed play. The RIR was 19.0% (4/21). Most (75.0%) reinjuries occurred before RTP. There was no significant relationship between BAMIC and TRTP, although TRTP was less for grade 0 injuries and greater TRTP for injury site "c," albeit with small effect sizes.

Conclusion: Activity-related HS and QD tears are common in American football college athletes, particularly in preseason and practice. Injured athletes missed about 3 to 4 games, regardless of BAMIC, with 1 in 5 athletes suffering a reinjury. There was no relationship between BAMIC or muscle group and TRTP.

Clinical relevance: This appraisal of BAMIC in college and American football athletes highlights the prevalence of muscle tears and the need for prevention. Although BAMIC lacked value in this study, larger studies are needed to evaluate BAMIC in this population.

Comparative analysis of treatment strategies for groin injuries in athletes: Effects on successful recovery and recurrence of sports conditions – A systematic review. Apunts Sports Medicine, 60: 100481, 2025.

Abstract - Core muscle injuries, characterized by pubic region pain from muscle and tendon damage, present challenges in athletic recovery. Treatment for pubic pain in athletes often involves a combination of clinical approaches and surgical interventions, typically requiring long recovery periods. This study systematically reviews the treatment methods for groin pain in athletes and evaluates the return to sports outcomes. A total of 86 studies were reviewed, with 30 included in the qualitative analysis. The findings suggest a stepwise approach starting with conservative treatments, such as shockwave therapy, manual therapy, and exercise. If no improvement is observed within 3 to 6 months,

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surgery is considered. Procedures like adductor tenotomy and selective release have shown effectiveness in facilitating return to sports. Core muscle injuries require personalized treatment plans, with promising outcomes from conservative methods and effective surgical options for severe cases, leading to safe and complete recovery in most athletes. Level of evidence: 1

Concurrent agreement and test-retest reliability of a Global Positioning System device for measuring maximal horizontal deceleration ability in elite youth academy soccer players. International Journal of Sports Physiology and Performance, Ahead of Print: 1-6, 2025.

Purpose: Investigate the concurrent agreement and test–retest reliability of 10-Hz global-positioning-system (GPS) device against a criterion measure (47-Hz radar device) to assess maximal horizontal deceleration ability (maximum deceleration [DECMax], average deceleration [DECAve], time to stop, and distance to stop).

Methods: Thirty-two male elite youth academy soccer players (age 18.1 [1.6] y, body mass 76.6 [7.9] kg) completed the acceleration–deceleration ability test with 16 completing a second test to assess test–retest reliability. Maximal horizontal deceleration ability was measured concurrently using GPSRaw (10-Hz data), GPSExport (STATSports software), and a radar device. Bland–Altman method and equivalence testing assessed concurrent agreement and intraclass correlations with coefficient of variation (%) was used to assess test–retest reliability.

Results: Equivalence testing showed mean difference between the radar device and GPS-derived values of DECAve and DECMax were within equivalence bounds. GPSRaw and GPSExport derived values of DECMax showed good overall (intraclass correlations = .84–.86, coefficient of variation % = 4.50–5.48) test–retest reliability.

Conclusion: Practitioners can consider using deceleration variables (DECAve and DECMax) obtained from GPS as a cost-effective, valid, and reliable alternative to radar technology to assess maximal horizontal deceleration ability in team-sport players.

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Distribution of position-specific head impact severities among professional and Division 1 collegiate American football athletes during games. BMJ Open Sport & Exercise Medicine, 11: e002365, 2025.

Objective: To compare the severity of head impacts between professional and Division I (D-I) collegiate football games for the purpose of improving protective equipment.

Methods: A total of 243 football players from the National Football League (NFL) and from D-I of the National Collegiate Athletic Association (NCAA) were equipped with instrumented mouthpieces capable of measuring six degrees-of-freedom head kinematics. Head impacts were processed using a custom algorithm and combined with game period descriptors to produce a curated dataset for analysis. Head impact severity distributions for several kinematic-based metrics were compared within position groupings between leagues.

Results: A total of 11 038 head impacts greater than 10 g from 1208 player-games were collected during 286 player-seasons (2019-2022). No significant differences were found between leagues in the distributions of kinematic-based metrics for all investigated position groupings ($p \geq 0.320$). The median and IQRs for peak linear acceleration for NFL and NCAA were 17.2 (9.3) g and 17.0 (8.6) g for linemen, 20.7 (13.8) g and 20.0 (13.5) g for hybrid and 21.0 (17.0) g and 20.8 (15.5) g for speed position groupings, respectively.

Conclusion: The absence of statistically significant differences in the distributions of head impact severity between professional and D-I collegiate football players indicates that these data can be combined for the purpose of understanding the range of loading conditions for which new protective equipment, such as position-specific helmets, should be designed. This observation underscores the potential for knowledge transfer regarding biomechanical factors affecting head loading across professional and D-I college football, highlighting crucial implications for innovation in protective equipment.

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Effects of protein supplementation during early rehabilitation on muscle volume and function after acute muscle strain injuries: A randomized controlled trial. Scandinavian Journal of Medicine & Science in Sports, 35: e70043, 2025.

Abstract - Muscle strain injuries are common in sports, with a high recurrence rate and loss of muscle mass. Whether protein supplementation can counteract the detrimental effects of strain injuries during rehabilitation has not been explored. We investigated the effects of protein supplementation during early rehabilitation of acute strain injuries on muscle volume and function. Fifty recreational athletes were enrolled for a double-blinded, randomized controlled trial comparing the effects of twice-daily whey protein supplementation to isocaloric placebo during early rehabilitation after acute hamstring or calf strain injuries. The primary outcome was changes in muscle volume during the 3 months of intervention, assessed by magnetic resonance imaging. Secondary outcomes included muscle strength, time to return to sport (RTS), and aponeurosis volume. Long-term changes were assessed 12 months after the injury. Muscle volume decreased in the injured muscle compared to the contralateral healthy muscle during the intervention [-9.4% (-13.5% to -5.3%), $p < 0.0001$], and persisted at 12 months. No effects of protein supplementation were observed. Further, there was no effect of protein supplementation on muscle strength or RTS. There was a persistent increase in aponeurosis volume associated with the injured muscle at 3 months [sixfold enlarged, +17.3 cm^3 (+8.3 to 26.3 cm^3), $p = 0.0005$] and 12 months [fivefold enlarged, +10.4 cm^3 (5.3-15.4 cm^3), $p = 0.0003$], but it was unaffected by protein supplementation. In conclusion, muscle strain injuries cause persistent atrophy of the injured musculature and enlargement of the muscle aponeurosis, which was not counteracted by protein supplementation during the rehabilitation period.

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Exploratory data analysis of NFL player performance and impact of age on speed and acceleration. 24th International Symposium INFOTEH-JAHORINA, 1-5, 2025.

Abstract—This study examines the role of speed and agility in shaping player performance and game strategies in American football. By investigating the relationship between speed, acceleration, and age, the study uncovers trends that shed light on what makes players effective on the field. A comprehensive data analysis process, including preprocessing, feature extraction, and normalization, ensures a robust examination of these relationships. The findings highlight the complex dynamics between physical attributes and age, revealing insights into player development and performance dynamics. While offering valuable insights, the study also emphasizes the need for further investigation to account for outliers and refine the understanding of these critical performance factors. This study lays a solid foundation for improving how we evaluate and predict player potential, making a valuable contribution to performance analysis in football.

High return to competition rate after on-field rehabilitation in competitive male soccer players after ACL reconstruction: GPS tracking in 100 consecutive cases. The Orthopaedic Journal of Sports Medicine, 13(3): 1-10, 2025.

Background: Despite published guidelines describing on-field rehabilitation (OFR) frameworks for soccer, available evidence for practitioners who work with players with anterior cruciate ligament reconstruction (ACLR) is limited.

Purpose: To document the activity and workloads completed by a large cohort of amateur and professional soccer players during OFR following ACLR after completing their indoor rehabilitation and to establish their return to competition (RTC) outcomes.

Study design: Case series; Level of evidence, 4.

Methods: OFR measurements/activities, global positioning system (GPS), and heart rate data were collected from 100 male 11-a-side soccer players with ACLR undergoing a criteria-based rehabilitation

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process, concluding with a 5-stage OFR program. Consent was obtained directly from the players involved in this study before completing a follow-up questionnaire to document RTC outcomes. Differences between the level of play (professional and amateur) and 5 OFR stages were investigated using separate linear mixed models.

Results: A minimum 9-month follow-up was possible for 97 players (97%), with a median time of 2.3 years after ACLR and 84% RTC, with higher rates in professionals (100%) than amateurs (80%). Ten (10%) players sustained an ACL reinjury. Professionals completed more OFR sessions (20.6 ± 7.7 vs 13.2 ± 7.7; $P < .001$) over a shorter period (44.7 ± 30.3 vs 59.3 ± 28.5 days; $P = .044$) and achieved higher workloads mostly in the high-intensity GPS metrics in each OFR stage. Typical external workload outputs in the final OFR stage aligned with team training demands for the total distance (TD) (106%), high-intensity distance (HID) (104%), peak speed (PS) (88%), acceleration distance (ACC) (110%), and deceleration distance (DEC) (48%), but they were lower compared with match play demands (TD: 44%; HID: 51%; PS: 82%; ACC: 63%; and DEC: 26%).

Conclusion: High RTC rates were reported in those players who participated in OFR after indoor rehabilitation. Completion of all five OFR stages almost prepared them for team training demands; however, workloads remain low compared to match play.

Impact of long-haul airline travel on athletic performance and recovery: A critical review of the literature. Experimental Physiology, Ahead of Print: 1-19, 2025.

Abstract - Participation in many important sport events (e.g., World championships, Olympics) requires athletes to fly >4 h and to cross several time zones. This transmeridian travel results in a transient desynchronization of the body's circadian rhythms due to a disconnect between the timing of the endogenous circadian oscillator and the external stimuli, manifested as 'jet lag'. While recent reviews highlight the importance of managing jet lag, the time required for resynchronization of the internal clock and dissipation of jet lag symptoms has not yet been summarized. Further, although the literature reports that rapid transmeridian travel is detrimental for athletes' performance, empirical evidence from studies involving athletes is equivocal. Herein, we summarize the evidence that the variability in

responses to transmeridian travel can be attributed to differences in (i) travel (real vs. simulated, westbound vs. eastbound, time zones crossed, during normal waking hours vs. normal sleep time), (ii) testing (assessment of performance vs. factors related to performance), and (iii) timing of the testing (destination time vs. 'body time'), and we offer the possibility that differences in (iv) teams, (v) traits, and (vi) tournaments may also be implicated. We focus on (i) aerobic power/endurance, (ii) anaerobic power and capacity, (iii) strength, and (iv) mood state, sleep quantity and quality, and jet lag symptoms in this literature review, which is limited to athletes or physically active participants, travelling west or east crossing four or more time zones.

Implementing the reverse acute to chronic workload ratio model to improve movement capacity and roster availability: An example using data from the NFL. International Journal of Strength and Conditioning, 5(1): 1-9, 2025.

Abstract - American football athletes require the development of workload capacity for repeated high-intensity efforts, and successful athletes are adept at accelerating, decelerating, and changing directions. The prescription of appropriate training volume stimulus can be difficult to determine, as there are very few guidelines for prescribing sport-specific acceleration, deceleration, and maximum velocity efforts. Preparatory training stimulus has to closely match in-game demands, but at the same time, practitioners need to avoid excessive workloads and undertraining to mitigate workload progression-related injuries and maximize roster availability. The acute-to-chronic workload ratio (ACWR) approach is based upon the fitness: fatigue ratio, which allows practitioners to monitor workloads. New technology allows for in-game positional tracking and these advancements are accessible to the public. By measuring in-game movement, coaches can quantify key metrics like the number of accelerations and average distance covered. These metrics provide a snapshot of in-game demands and performance requirements. Using a reverse engineering approach, coaches can utilize ACWRs to calculate predefined targets to ensure athletes are adequately prepared for gameplay. Here we use the ACWR concept and previously reported in-game data derived from the National Football League to show how to reverse engineer the targeted number of efforts and distances to assist in preparatory pre-season training program design. This approach, which we term the Reverse ACWR Method, can be used to set guidelines for training volumes and workload progressions and provides a

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systematic, quantitative approach that complements periodization. As such, the Reverse ACWR Method allows practitioners to calculate target sport-specific workloads and training progressions derived from scientific-grounded methodology, which may enhance performance, readiness, and roster availability. Although this paper presents an example of how to use positional in-game data to prescribe American football training workloads, this model can be applied to any sport and team that has access to positional in-game movement data.

Influence of resistance training variables and the Nordic hamstring exercise on biceps femoris architectural adaptations in soccer players: A systematic review. Sports Health, Ahead of Print: 1-10, 2025.

Context: Manipulation of resistance training variables influences the structural and functional adaptations of muscle, having a great impact on sport performance and hamstring injury prevention.

Objective: To analyze how the main resistance training variables affect the biceps femoris long head architecture in soccer players.

Data sources: Five databases were searched from inception to January 2024.

Study selection: Studies that included training intervention groups and measured muscle architecture adaptations before and after the training program in soccer players were included.

Study design: Systematic review with meta-analysis.

Level of evidence: Level 2.

Data extraction: Muscle thickness, fascicle length, and pennation angle were extracted from included studies as main outcomes.

Results: Six studies and 12 training groups (168 participants) were analyzed. The effects of Nordic hamstring exercise (NHE) against soccer interventions, volume of training, and frequency of training as independent variables were analyzed. NHE significantly improved biceps femoris long head fascicle length ($P = 0.01$). Training twice a week did not show significant differences compared with training

once a week. Higher volumes of training (ie, >290 repetitions) in a period of 6 to 12 weeks with 57 repetitions per week demonstrated significant effects.

Conclusion: NHE lengthens the fascicle, especially if a sufficient volume (ie, >290 repetitions) and 2 days per week are performed. It is still unknown how the programming of some fundamental variables such as intensity, degree of effort, or exercise selection affects the muscle architecture of the biceps femoris long head.

Injury scenarios of ankle sprains in the National Football League: External rotation and eversion mechanisms. The American Journal of Sports Medicine, Ahead of Print: 1-10, 2025.

Background: Ankle sprains in American football resulting from external rotation and eversion (EE) mechanisms remain a persistent injury that often involves player-to-player contact. Currently, there is little research that describes the frequency or source of this contact in American football and what scenarios are responsible for these injuries across different player positions.

Purpose: To identify injury scenarios of ankle sprains that result from EE mechanisms in the National Football League (NFL).

Study design: Descriptive epidemiology study.

Methods: Ankle sprains reported during NFL games over 4 consecutive seasons (2017-2020) were identified through a query of the NFL Electronic Medical Record by an independent epidemiological company based on a set of clinical impression codes. Injuries with available video to view the injury enabled the identification of ankle sprains that resulted from EE mechanisms. Standardized terminology was developed to systematically describe and categorize each injury.

Results: A total of 257 ankle sprains resulting from EE mechanisms were identified from a review of 670 ankle sprains and grouped into 8 standardized injury scenarios. Direct contact to the injured player's foot, ankle, or leg from a large external mass, such as an opponent's pelvis, accounted for 79% of reviewed injuries. Direct contact from a large external mass was the most common scenario for all

positions except safeties, and 83% of running back injuries were the result of a tackle. Only 15% of injuries reviewed resulted from injury scenarios in which injurious loading was transmitted solely through the cleat-surface interaction. Safeties and cornerbacks were more commonly involved in these nondirect injury scenarios, such as change of direction, where current interventions may prove more effective.

Conclusion: EE ankle sprains were observed to occur from 8 different injury scenarios that varied by player position. Interventions through training, bracing, or changing cleat-surface interaction may not be effective for all injury scenarios. Tackling technique may be a unique intervention for running backs.

Introducing the Novel Deceleration Threshold Method: Comparative reliability to previous sprint deceleration analysis methods in team-sport athletes. European Journal of Sport Science, 25: e12278, 2025.

Abstract - Pre-planned deceleration assessments are vital for profiling team sport athletes as they provide a measure of the athletes' ability to implement effective performance and protection strategies to tolerate the substantial mechanical forces. Although task requirements of these linear assessments are generally consistent, current research employs low sampling technology, which limits the ability to define the deceleration start point and subsequently the reliability. Therefore, this study compared the intra- and inter-day reliability of three deceleration assessment analysis methods: set distance, peak velocity, and the novel deceleration threshold, using a motorized linear encoder device. Ten female and 10 male subjects performed three maximal effort 30 m sprint deceleration trials over two testing days. Each trial was filtered and analyzed using a customized code to calculate key deceleration metrics: distance-to-stop (DTS), time-to-stop (TTS), and average deceleration (DEC_{ave}) for each analysis method. Intraclass correlation coefficients (ICC) using two-way fixed effect models, coefficients of variation (CV%), and standard errors of measurement (SEM) were calculated to determine the intra-day reliability across all three trials of the first testing day, and inter-day reliability using the: best, average of best two, and average of all trials. The deceleration threshold method using the average of the best two trials from each testing day, exhibited excellent inter-day reliability for the key metrics (TTS: ICC = 0.93, CV% = 6.4; DTS: ICC = 0.97, CV% = 5.3; DEC_{ave}: ICC = 0.92, CV% = 7.1). To optimize reliability and sensitivity of

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detecting a meaningful change in sprint deceleration performance practitioners should consider using the average of two best trials analyzed with the deceleration threshold.

Is it all in your head? Placebo effects in concussion prevention. Sports Medicine, Ahead of Print: 1-17, 2025.

Abstract - Sports-related concussions (SRCs) are a major health issue in athletes of varying ages, ability levels, and sports. Concerns over the short- and long-term consequences of SRCs have incentivized a wealth of products and policies aimed at reducing SRC risk. Research suggesting the effectiveness of such interventions at reducing SRCs has facilitated their adoption by sports organizations and, in some cases, product commercialization. However, the body of SRC mitigation research is almost entirely devoid of placebo or sham groups, which raises important questions about the true clinical effectiveness of these interventions. This Current Opinion explores the plausibility of placebo effects within the scope of SRC prevention, describes why the lack of placebo/sham groups in the current body of literature is problematic, and provides recommendations for including placebo/sham groups in future SRC research.

Less pain, same gain? A randomized controlled trial comparing assisted versus non-assisted Nordic hamstring exercise training in elite youth football players. Physical Therapy in Sport, 73: 39-47, 2025.

Introduction: The Nordic Hamstring exercise (NHE) increases hamstring strength, but muscle soreness and excessive fatigue may hinder adoption. The assisted NHE (A-NHE), with an elastic band around the chest lowering the force in the eccentric lowering phase, may be a less demanding alternative. We investigated the effectiveness of A-NHE versus NHE on hamstring strength, muscle soreness, and perceived exertion.

Methods: We randomized 31 youth elite football players (under-14 and -15) from a single professional football academy to either NHE or A-NHE, who performed the same supervised 8-week program. The A-NHE was performed with an elastic band, progressing from heavy to light assistance. A blinded tester

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assessed Nordic hamstring strength, while hamstring muscle soreness and perceived exertion were assessed during each session.

Results: Intention-to-treat analysis showed no group by time interaction ($p = 0.61$) for Nordic hamstring strength. Both groups improved significantly (A-NHE mean difference: 3.7 kg, 95% CI [2.1; 5.3]; NHE mean difference: 3.2 kg, 95% CI [1.2; 5.2]) corresponding to approximately 20%. The NHE group reported significantly higher hamstring muscle soreness (beta = 1.77, $p < 0.01$) and perceived exertion (beta = 1.54, $p < 0.01$).

Conclusion: Eight weeks of assisted-versus regular Nordic Hamstring training led to similar strength gains, but with lower hamstring muscle soreness and perceived exertion.

National Football League players have higher rates of knee extensor mechanism tears during short and normal rest weeks versus long rest weeks. The Journal of Knee Surgery, Ahead of Print: 1-17, 2025.

Background: Lower extremity injuries, particularly knee injuries, in National Football League (NFL) players are common. Extensor mechanism ruptures often require season-ending surgery. Research on NFL injury risk factors, including time between games, is growing, but studies on rest periods and knee injuries have shown mixed results. No prior research has specifically examined extensor mechanism tears in relation to short, normal, or long rest periods between games.

Methods: An online search was conducted to identify NFL players who tore their quadriceps tendon or patellar tendon between the 2009-2023 seasons. Only regular season injuries after week 1 were included. The date of injury and prior game date were collected in order to calculate the length of rest. Next, each injury was categorized as a short (<7 days), normal (7 days), or long (>7 days) week injury. Other factors, such as age, football position, and playing surface (natural grass or artificial turf) were also collected.

Results: A total of 58 players met inclusion criteria. There was no difference in extensor mechanism injury rates between short weeks and normal weeks. Players were 4.7 times more likely to tear their extensor mechanism during normal weeks than long weeks ($p = 0.0039$) and 3.7 times more likely to

tear their extensor mechanism during short weeks than long weeks (p = 0.0385). Regarding playing surface, extensor mechanism tears were more likely to occur on artificial turf than natural grass (p = 0.0324).

Conclusion: During the 2009-2023 NFL seasons, players were found to have 4.7 times higher rates of knee extensor mechanism tears during short (< 7 days) rest weeks and 3.7 times higher rates during normal (7 days) rest weeks between games, when compared to long (>7 days) rest weeks. There was no difference in knee extensor mechanism injury rates between the short and normal rest weeks.

Nutritional strategies to improve post-exercise recovery and subsequent exercise performance: A narrative review. Sports Medicine, Ahead of Print: 1-19, 2025.

Abstract - Post-exercise recovery strategies influence the body's ability to restore physiological homeostasis, replenish energy stores, repair muscle damage, and promote desired adaptations, which improve exercise performance. This narrative review examines the impact of nutritional strategies commonly used for enhancing recovery and subsequent exercise performance, particularly when athletes face short recovery periods. Carbohydrate ingestion is essential for glycogen replenishment, especially within the initial hours post-exercise, with its impact dependent on the types, timing, and amount. Protein is essential for accelerating muscle recovery and achieving a positive nitrogen balance, depending on the type and dosage. The co-ingestion of carbohydrates with proteins or fats is explored for its role in maximizing glycogen resynthesis and muscle repair, with evidence supporting the addition of protein to suboptimal carbohydrate intake for enhanced recovery. Moreover, this review addresses the potential benefits of creatine and caffeine co-ingestion for accelerating glycogen synthesis and improving subsequent performance. Hydration strategies, including the use of milk-based beverages and electrolyte solutions, are also discussed, emphasizing their importance in maintaining fluid balance and optimizing recovery. This review also highlights the emerging role of micronutrients such as omega-3 fatty acids, antioxidants, and sodium bicarbonate in reducing muscle damage and improving acid-base balance. Evidence supports the tailored use of these nutritional strategies, particularly for athletes managing tight competition/training schedules. Future research should focus on refining individualized

approaches for recovery and investigating the impact of novel supplements on subsequent performance.

Post-operative rehabilitation and return to sport criteria following distal biceps tendon rupture surgery. JSES International, Ahead of Print: 1-22, 2025.

Background: Within the clinical literature, there is no consensus on optimal rehabilitation for return to sport (RTS) after distal biceps tendon rupture (DBTR) repair. The authors hypothesize that surgeons will utilize time-based criteria for RTS rather than performance-based or functional criteria for RTS.

Methods: Level 1-4 studies that evaluated DBTR with a minimum of 12 months follow-up were eligible. Exclusion criteria included studies that treated DBTR nonoperatively, lacked RTS criteria, or included revision surgery. Studies were evaluated for RTS criteria, timeline/rate of RTS, and methodologic quality.

Results: Out of 671 studies identified, 5 met inclusion criteria. These included 268 patients playing 22 different sports. Timeline for return ranged from 3 to 130 weeks. Range of motion exercises followed by strengthening (100%) and multiphase rehabilitation program (100%) were the most common parameters emphasized in rehabilitation protocols, followed by immediate postoperative cast/splint immobilization (80%). All studies included a timeline for RTS; however, no article listed subjective or specific objective measurement criteria for RTS.

Conclusion: There is no consensus on RTS criteria in the literature. Eighty percent of the included studies reported utilizing initial cast/splint immobilization; 100% reported a multiphase program consisting of some form of progressive range of motion followed by strengthening. Sixty percent reported a supervised physical therapy routine. One hundred percent reported timing after surgery as a criterion for RTS, but this timing varied greatly across the included studies.

Level of evidence: Level IV Systematic Review

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Practices and perceptions in hamstring training for injury risk mitigation: A survey-based mixed-methods analysis. International Journal of Sports Science & Coaching, Ahead of Print: 1-17, 2025.

Abstract - In this study, the practices and perceptions of sport and exercise practitioners regarding hamstring training for injury risk mitigation were investigated. Hamstring strain injuries (HSIs) remain a prevalent issue across sports, despite extensive research. An anonymous survey was conducted to assess how practitioners integrate evidence-based strategies, such as the Nordic hamstring exercise (NHE) and high-speed running (HSR), into their training programs. The survey included both fixed-response and open-ended questions, enabling a mixed-methods analysis of training approaches. A total of 47 practitioners responded, revealing that most incorporate the NHE and HSR into both in-season and off-season training. However, there were notable differences in volume, with higher HSR distances and NHE volumes programmed in the off-season. Despite widespread adoption of the NHE, practitioners indicated that the volumes of NHE used are still likely higher than the smallest effective dosages reported in the literature. Thematic analyses highlighted several key challenges practitioners face, including athlete compliance with training programs, multi-disciplinary team coordination, and time limitations. Practitioners emphasized the importance of exposing athletes to maximal running efforts to reduce HSI risk, while also integrating strength training exercises that target the hamstrings. The study serves to highlight a need for further improvements in communication among teams and better athlete education which could enhance the effectiveness of hamstring training programs. Additionally, the findings of the current study indicate a need for future studies that explore the combined effects of resistance training and HSR, as well as how scientific recommendations can be more effectively implemented in practical settings.

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Predicting non-contact injuries of professional football players using machine learning. PLOS ONE, 20(1): e0315481, 2025.

Abstract - Noncontact injuries are prevalent among professional football players. Yet, most research on this topic is retrospective, focusing solely on statistical correlations between Global Positioning System (GPS) metrics and injury occurrence, overlooking the multifactorial nature of injuries. This study introduces an automated injury identification and prediction approach using machine learning, leveraging GPS data and player-specific parameters. A sample of 34 male professional players from a Portuguese first-division team was analyzed, combining GPS data from Catapult receivers with descriptive variables for machine learning models— Support Vector Machines (SVMs), Feedforward Neural Networks (FNNs), and Adaptive Boosting (AdaBoost)—to predict injuries. These models, particularly the SVMs with cost sensitive learning, showed high accuracy in detecting injury events, achieving a sensitivity of 71.43%, specificity of 74.19%, and overall accuracy of 74.22%. Key predictive factors included the player's position, session type, player load, velocity and acceleration. The developed models are notable for their balanced sensitivity and specificity, efficiency without extensive manual data collection, and capability to predict injuries for short time frames. These advancements will aid coaching staff in identifying high-risk players, optimizing team performance, and reducing rehabilitation costs.

Quantifying leg muscle disuse atrophy during bed rest using DXA, CT, and MRI. European Journal of Sport Science, 25: e122999, 2025.

Abstract - This study evaluated whether dual-energy X-ray absorptiometry (DXA), computed tomography (CT), and magnetic resonance imaging (MRI) provide comparable outcomes in quantifying disuse-induced skeletal muscle atrophy. Although the calculation of muscle volume using MRI analysis may be considered the gold standard, the method remains labor intense and, as such, less practical and more costly. In this context, we also evaluated the efficacy of a commercially available automated MRI analysis method to measure changes in leg muscle volume after two weeks of bed rest. Twelve healthy, male adults (age: 24 ± 3 years, BMI: 23.7 ± 3.1 kg/m^2) were subjected to 2 weeks of strict bed rest. Leg muscle assessments were performed before and after bed rest using DXA, single slice (thigh) CT, and

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MRI. MRI data analyses were performed using both a manual and automated (AMRA) method. Leg lean mass, as assessed with DXA, declined by 5% following bed rest (from 10.2 ± 1.6 to 9.7 ± 1.6 kg; $p < 0.001$). The thigh muscle cross-sectional area, as assessed with CT, declined by 6% following bed rest (from 155 ± 26 to 146 ± 24 cm^2; $p < 0.001$). Muscle volume, as assessed using MRI, declined by 5% following bed rest, both when assessed manually (from 7.1 ± 1.1 to 6.7 ± 1.0 L; $p < 0.001$) and automatically (from 7.2 ± 1.1 to 6.8 ± 1.0 L; $p < 0.001$). A very strong correlation ($r = 0.96$; $p < 0.001$) with a low bias (-0.11 ± 0.29 L) was observed between manual and automated muscle volume analysis. DXA, CT, and MRI all show a ~5% decline in leg muscle quantity following two weeks of bed rest in healthy adults. When using MRI, disuse muscle atrophy can be accurately quantified using an automated approach, rendering time-consuming manual analysis obsolete.

Reconsidering exercise selection with EMG: Poor agreement between ranking hip exercises with gluteal EMG and muscle force. Medicine and Science in Sports and Exercise, Ahead of Print: 1-49, 2025.

Purpose: Electromyography (EMG) is commonly used to inform exercise selection for injury prevention and rehabilitation. However, mechanical tension is more important for driving adaptation than muscle activation alone. Therefore, this study aimed to investigate the effect of ranking hip-focused exercises based on the magnitude of gluteal surface EMG compared to estimates of muscle forces and the implications for exercise selection.

Methods: Fourteen healthy female footballers (18-30 years old) performed eight hip-focused exercises using body weight and 12-repetition maximum loads. Full body kinematics, ground reaction forces, and surface EMG from 12 lower limb muscles were recorded and used as inputs to a neuromusculoskeletal model to estimate gluteal muscle forces. Normalized EMG and normalized muscle force rankings were compared using Spearman's correlations and the linear relationship between normalized EMG and muscle force was compared using linear mixed effects models.

Results: There was a weak relationship between exercise rankings based on EMG and muscle forces (Spearman's $\rho = 0.29$-0.51). Peak EMG amplitude alone explained 5% of gluteus maximus peak muscle force ($R2 = 0.05$) and 19% of gluteus medius peak muscle force ($R2 = 0.19$). However, when accounting

for exercise and participant sources of variation, peak EMG amplitude explained 80-85% of gluteus maximus and medius peak normalized muscle force.

Conclusions: Ranking gluteal resistance exercises by EMG amplitude resulted in a different order to ranking exercises by estimated muscle forces. For exercise selection, EMG may only be useful when comparing within an individual and between biomechanically similar exercises. Caution is warranted when basing exercise selection on gluteal EMG amplitude alone.

Reference values for hydration biomarkers: Optimizing athletic performance and recovery. Open Access Journal of Sports Medicine, 16: 31-50, 2025.

Abstract - The negative effects of dehydration or overhydration on exercise performance and health are widely recognized. However, the interindividual variability of fluid imbalances among athletes and across various sports is large, due to the complex interactions of physiological, environmental, and sport-specific factors. Such complexity not only makes it difficult to predict fluid needs prior to competition or training sessions, but also supports the creation of an individualized hydration plan (IHP) for each athlete. Measurements of valid, field-expedient biomarkers such as body mass change, urine concentration, and thirst enable ongoing monitoring of an athlete's hydration state and are integral components of an IHP. Unfortunately, no extensive repository of sport-relevant hydration biomarker data exists. Therefore, this narrative review presents a novel inventory of pre- and post-exercise reference values for body mass change, urine specific gravity, and subjective rating of thirst. These reference values were identified via electronic database searches that discovered field studies of competitive events, weight category sports, training sessions, and routine daily activities. We propose that comparing an athlete's real-time body mass change, urine specific gravity, and thirst rating to previously published reference values will clarify the extent of dehydration or overhydration, guide rehydration efforts, and optimize subsequent exercise performance, recovery, and health.

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Relative rest correlates to winning percentage in 2022 National Football League season: A preliminary study. International Journal of Strength and Conditioning, 5(1): 1-6, 2025.

Abstract - The purpose of this investigation was to examine whether season average relative rest (RR) influences winning percentage (WIN%) in professional American Football. We define RR as the difference in rest intervals between teams from prior games. To address this, we analyzed each game played in the 2022 NFL season (n=271) and calculated rest intervals between each successive game for each team. The rest interval was defined as the number of days between the schedule start for each successive game, with all teams being assigned a rest interval of 0 during Week 1. Each team's RR was then determined by subtracting the opposing team's rest interval. The RR for each game for all teams were then averaged across the season. Subsequently, a correlation between season average RR and WIN% was performed and a tertial split in season average RR was conducted to separate teams into HIGH (teams 1-11), MID (teams 12-21) and LOW (teams 22-32). A one-way ANOVA comparing WIN% between HIGH, MID and LOW was then conducted while independent samples t-tests were employed to assess season average RR between teams with records above (WIN) and below 0.500 (LOSE) and those qualifying (PO) and not qualifying (NPO) for the playoffs. Data indicate that a significant positive relationship exists between WIN% and season average RR ($r=0.338$; $p = 0.029$). Moreover, successful teams (WIN/PO) had significantly greater season average RR than unsuccessful teams (LOSE/NPO). Therefore, a team's rest interval, relative to their opponent was associated with the outcome of the competition during the 2022 NFL season.

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Sprint running mechanics are associated with hamstring strain injury: A 6-month prospective cohort study of 126 elite male footballers. British Journal of Sports Medicine, Ahead of Print: 1-8, 2025.

Objective: To investigate the association between sprint running biomechanics and sprint-related hamstring strain injury (HSI) in elite male football players.

Methods: This prospective cohort study recruited 126 professional male football players from eight clubs in the English football league, who were followed across a 6-month period. Maximal velocity sprint running videos (240 fps) were collected from five teams during preseason (June to August) and three teams during the in-season period (October to March) and subsequently assessed using the Sprint Mechanics Assessment Score (S-MAS) by a single, blinded assessor. Sprint-related HSI within the previous 12 months and any new MRI-confirmed sprint-related HSI were reported by club medical staff. Incidence rate ratios were calculated using a Poisson regression model to determine the association between S-MAS and new sprint-related HSIs.

Results: There were 23 players with a previous sprint-related HSI and 17 new HSIs during the follow-up period, with 14 sprint-related injuries. S-MAS values were significantly greater among players with a previous HSI (median difference (MD): 1, p=0.007, 95% CI: 0 to 3) and those sustaining a new sprint-related HSI (MD: 2, p=0.006, 95% CI: 1 to 3) compared with uninjured players. Adjusting for age and previous injury found a significant association between the S-MAS and prospective sprint-related HSIs, with an adjusted incidence rate ratio of 1.33 (95% CI: 1.01 to 1.76) for each one-point increase in S-MAS.

Conclusions: This is the first study to identify an association between sprint running kinematics and prospective sprint-related HSI in elite male football players. Sprint running mechanics assessed using the S-MAS were associated with both past and future HSIs, with a 33% increase in the risk of a new HSI with every one-point increase in S-MAS. Given the association to injury, evaluating sprint mechanics within rehabilitation and injury prevention may be warranted.

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Use of Ankle-Go to assess and predict return to sport after lateral ankle reconstruction for chronic ankle instability. The Orthopaedic Journal of Sports Medicine, 13(3): 1-9, 2025.

Background: Chronic ankle instability (CAI) develops in 20% to 40% of patients after ankle sprains because of inappropriate management or a premature return to sports (RTS). Patients become candidates for surgery when nonoperative treatment fails. The results of arthroscopic lateral ligament reconstruction have been shown to be excellent. However, there is no consensus on the RTS criteria in these cases. The Ankle-GO score, which has been validated in lateral ankle sprains, is an evidence-based protocol for RTS.

Purpose: To evaluate the discriminant and predictive capacity of the Ankle-GO score for RTS after arthroscopic lateral ankle reconstruction.

Study design: Cohort study. Level of evidence, 2.

Methods: This prospective multicenter cohort study, conducted between January 2022 and January 2023, evaluated patients with CAI who underwent arthroscopic anatomic lateral ankle ligament reconstruction using gracilis tendon autografts or allografts. The RTS rates at 4 and 6 months postoperatively were the primary and secondary outcomes, respectively. The evaluated variable at both timelines was the Ankle-GO score-a composite of 4 functional tests and 3 questionnaires. The study determined the discriminant validity of the Ankle-GO score for RTS at each timeline-Ankle-GO scores at 4 months for RTS at 4 months; Ankle-GO scores at 6 months for RTS at 6 months. The predictive capacity of the Ankle-GO score at 4 months for RTS at 6 months was also evaluated.

Results: A total of 51 patients, with a mean age of 32.8 years, were included, of whom 23 (45%) had returned to sports at 4 months and 39 (76.5%) at 6 months. The mean Ankle-GO score was significantly higher in the group that returned to sports than in the group that did not (12 ± 4.7 vs 8.8 ± 4.7 at 4 months, respectively, $P = .003$; 16.8 ± 3.6 vs 11.3 ± 4.8 at 6 months, respectively, $P = .002$).The predictive capacity of the Ankle-GO score at 4 months for RTS at 6 months was good, with an area under the curve of 0.74 (95% CI, 0.58-0.90; $P = .005$). The optimal threshold was 6 (sensitivity, 95%; specificity, 50%). The odds ratio of RTS when surpassing the Ankle-GO cutoff score was 18.5 (95% CI, 3-113.95; $P = .002$).

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Conclusion: The Ankle-GO score is effective in discriminating the RTS status at 4 and 6 months after lateral ligament reconstruction of the ankle. In addition, the Ankle-GO score at 4 months was also moderately predictive of RTS at 6 months after surgery.

Validation of a composite outcome score for assessing return to sports after Achilles tendon repair. The American Journal of Sports Medicine, Ahead of Print: 1-9, 2025.

Background: The Ankle-GO is a composite outcome score based on 4 functional tests and 2 patient-reported outcome measures. It was initially described and validated for its ability to predict return to sports (RTS) at the same level of play after lateral ankle sprains.

Purpose/hypothesis: The main aim of this study was to assess the psychometric properties of the Ankle-GO in patients after Achilles tendon repair. The second objective was to evaluate its ability to predict RTS at the same level of play. It was hypothesized that this composite outcome score is a valid and reliable tool that is able to discriminate and predict the level of RTS at 9 months after Achilles tendon repair.

Study design: Case-control study; Level of evidence, 3.

Methods: This was a prospective multicenter study. Patients who were active in sports and underwent primary acute Achilles tendon repair between April 2021 and December 2022 were included. A control group comprising 30 participants with no history of lower limb injuries was also evaluated. After standard rehabilitation for the first 2 months, the Ankle-GO was administered at 6 and 9 months after repair by the same experienced physical therapist. During testing, the patients were asked whether they were able to practice their main sport again and, if so, whether at the same or a lower level of play. A comparison between control participants and patients at 9 months after surgery was performed.

Results: A total of 50 patients (24 women and 26 men), with a mean age of 38.3 ± 10.1 years, were included. No patients were lost to follow-up at 9 months. The Ankle-GO score at 6 months was 10.7 ± 4.8. At 9 months, the score had risen by 4.8 points. All participants scored above the minimum threshold. The minimum score achieved was 3 points (n = 5 [10%]). Only 1 patient (2%) obtained the

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maximum score (25 points). There was a significant difference in the Ankle-GO score between patients at 6 and 9 months and between control participants and patients at 9 months after Achilles tendon repair. Of the 50 patients included in the study, 14 (28%) returned to their preinjury level of sport at 9 months. The ability of the Ankle-GO score at 6 months to predict RTS at the preinjury level at 9 months was good (area under the curve = 0.71 [95% CI, 0.56-0.85]; $P < .01$). A Youden index of 0.42 was observed for a cut-off score of 8 points, which corresponds to a sensitivity of 81.8% and a specificity of 41.2%.

Conclusion: The Ankle-GO is a valid tool to evaluate and discriminate patients during the RTS continuum after Achilles tendon repair and to predict RTS at the same level at 9 months after surgery. This composite outcome score assesses ankle function and the patient's psychological readiness to avoid premature RTS and worsening of the injury. Further studies are needed to assess its value in identifying and predicting patients at risk of a recurrence.